

CSP Design Rule for Ass'y

기술그룹

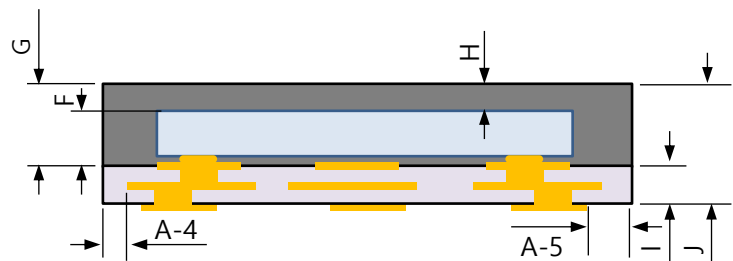
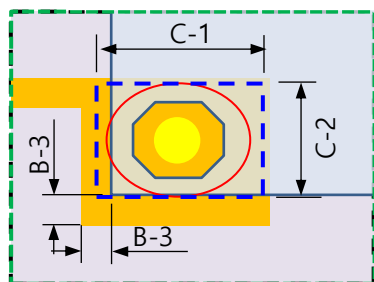
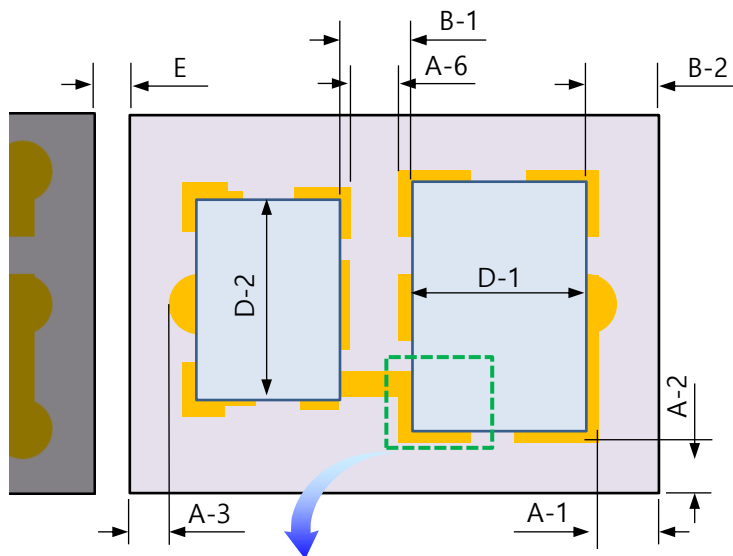
25.08.13

■ Revise History

Date: 25.08.13

Rev. No.	Contents	Date	a Drafter
00	Initial Release	17.07.26	YS.Shim
01	B-3, E, G, I, J	18.03.26	YS.Shim
02	A-3, F, c	18.10.19	YS.Shim
03	A-5, B-2, B-2, C-1 / C-2, D-1 / D-2, a-1 / a-2, a-1 / a-2, c, d-1 / d-2, d-3, e-1 / e-2, g	19.07.26	MJ.Kim
04	A-1 / A-2, B-3, b-1 / b-2, d-3	19.10.07	MJ.Kim
05	B-1, Changed Chip to Chip: 90um B-2, Changed Edge to Chip (소형화): Adv. 100um D-1 / D-2, Changed Chip Size: Max. 1,500um, 1,100um, Min. 400um	20.08.04	YS.Shim
06	J, Removed PKG Thickness a-1 / a-2, c-1 / c-2, f, Added for HS SAW c-2, Added Epoxy Margin e-1 / e-2, Added New Bump Margin 170 x 150um g, Changed PSV Opening	23.04.28	YS.Shim
07	a-1, a-2 Changed Chip Edge to Bump Pad (After Dicing 15um) f. Changed Chip Edge to 2 nd Pattern (After Dicing 17.5um) 55um Bump : 0907 HTCC, 0907 ~ 1612 PCB (NS, TS)	23.09.22	YS.Shim
08	기판 평탄도, Epoxy 침투 방지, Rule 추가 비대칭 Bump 설계 (0907, 1109, HTCC, 0907 ~ 1612 PCB) D-1, D2 Changed Max Chip Size b-1. Changed Bump Pad to Pad Gap → Bump Margin Gap Deleted 240umT Epoxy Film (방열)	24.06.26	YS.Shim
09	Added Al Pad Design (for NS, TS)	24.11.07	JO.Jo
10	Changed Bump Pad to Via Overlap Added Bump Margin Gap Max. Changed PSV (SiN) Opening Size	25.06.03	YS.Shim
11	1411 Design 55um Bump Chip Size Rule Remove Probe Pad	25.08.12	YS.Shim

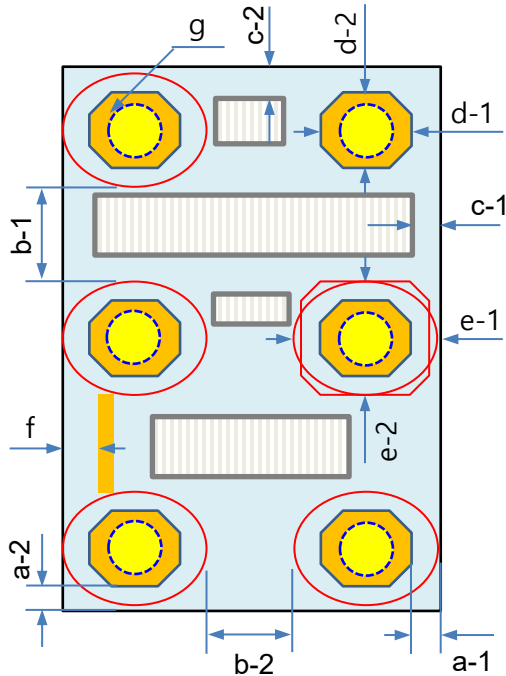
Substrate Design Rule for Ass'y



Items		Symbol	Specification - Unit : um			Related Process
			Adv.	Std.	Max.	
Edge to Fiducial (Top)		A-1 A-2	70	105	-	PKG Dicing / Molding
Edge to Pattern	Top	A-3	105	150	-	
	Inner	A-4	75	-	-	
Edge to BTM Pad		A-5	50	75	-	PKG Dicing / Test
Pattern to Pattern		A-6	50	60	-	업체 Rule 반영
Chip to Chip		B-1	-	90	-	Flip Bonding
Edge to Chip	1612 이하	B-2	100	175	-	PKG Dicing / Molding
	1814 이상		150			
Chip Outline to Fiducial		B-3	30	50	-	Flip Bonding
Substrate Bond Min. Pad Size	소형화 (55um)	C-1	180	260	-	Flip Bonding
		C-2	150	230	-	
	표준 (65/70um)	C-1	220	300	-	
		C-2	150	230	-	
	LNA (Plating)	C-1	140	-	-	
		C-2	140	-	-	
Chip Size (After Dicing)	HTCC	D-1	400	-	1,400	Wafer Dicing / Flip Bonding
		D-2	550	-	1,000	
	PCB	D-1	400	-	1,700	
		D-2	550	-	1,300	
PKG Dicing Width		E	80	150	-	PKG Dicing
Chip Thickness		F	150 ± 10 200 ± 10			Should be use same In PKG
Epoxy Film Thickness (RAW Material Height)		G	230 ± 10 260 ± 10			
Chip to Top Mold Gap		H	Min. 100			Marking
Substrate Thickness		I	150, Max. 170		180 ± 40	HTCC
			200 ± 40		255 ± 50	
			224 ± 20		242 ± 24	PCB
			320 ± 30		-	

- After Dicing Chip Size: (X – Blade 폭) x (Y – Blade 폭) || Step Cut시 Min. Blade 폭
- HTCC: Bump Pad - Via 겹침 설계 불가 / PCB: Bump on Via 설계
- * 모바일 기종으로 HTCC기판 재 사용시 Pad Center 기준 $\Phi 60\mu\text{m}$ 영역 Via 겹침 없을 것⁴⁾
- 전장용 기종 Grade2 : 65/70um 적용 (0907, 1109(A0, A4) 55um 허용)
- Top Pattern으로 상하 좌우가 구분 설계 필요 (FA분석 용이)
- 기판 평탄도 개선을 위해 Bump 아래 Layer Metal Pad 추가 설계 ¹⁾
- Epoxy 침투 방지를 위해 Top Layer, Pad 확장 or Dummy Pad 추가 설계 ²⁾
- ※ 소형화(55um) – 0907, 1109 (A0, A4), 1411 DPX HTCC, 0907 ~ 1612 PCB (대칭)
- HTCC 내층 Coat 적용 필수 기종 : 1411 DPX, 1109 Dual Filter

■ Chip Design Rule for Ass'y



* Bump UBM

Au	(0.215μm)
Ti	(0.240μm)
AlCu 0.5 or 8%	(2.00μm)
Ti	(0.055μm)

< Au Pad >

AlCu 0.5%	(1.700μm)
Ti	(0.400μm)
AlCu 0.5%	(0.56μm)
Ti	(0.033μm)

< Al Pad >

Items		Symbol	Specification - Unit : μm			Process Condition
			Adv.	Std.	Max.	
Chip Outline to Bump Pad Edge	NS / TS	a-1	32.5	-	-	Blade폭: 30
	NS	a-2	40	-	-	Blade폭: 50
	HS*, Bonded		-	50	-	Blade폭: 50/30
Bump Margin Gap		b-1	70	100	600	-
		b-2	70	100	680	-
Chip Outline to Reflector	NS / TS	c-1	-	47.5	-	Blade폭: 30
	NS		-	55	-	Blade폭: 50
	HS*, Bonded		-	55	-	Blade폭: 50/30
Chip Outline to IDT	NS(소형화) / TS	c-2	-	62.5	-	Blade폭: 30
	NS		-	70	-	Blade폭: 50
	HS*, Bonded		-	70	-	Blade폭: 50/30
Bump Pad Size	소형화 (55um)	d-1	-	110	-	-
		d-2	-	80	-	-
	표준 (65 / 70um)	d-1	-	130	-	-
		d-2	-	100	-	-
	LNA (Plating)	d-1	70	80	-	-
		d-2	70	80	-	-
Bump Pad Margin(Should not be pattern in this area)	소형화 (55um)	e-1	-	150	-	타원형
		e-2	-	120	-	-
	표준1 (65/70um)	e-1	-	190	-	타원형
		e-2	-	150	-	-
	표준2 (PCB, HTCC A0, A4)	e-1	-	170 (130)	Chamfer 20	8각형 (표준1, 2 혼용)
		e-2	-	150 (120)	Chamfer 15	-
Chip Outline to Pattern	NS(소형화)	f	-	32.5	-	Blade폭: 30
	NS		-	40	-	Blade폭: 50
	TS		-	35	-	Blade폭: 30
	HS*, Bonded		-	50	-	Blade폭: 50/30
PSV (SiN) Opening Size	소형화 (55um)	g	-	60	-	-
	표준 (65um)		-	80	-	-
	표준 (70um)		-	70	-	-
			-	80	-	기술그룹 협의 필요

- Chip Size : Before Dicing 기준

- Bump 비대칭 디자인 기술그룹 협의 (0907, 1109(A0, A4) HTCC, 0907 ~ 1612 PCB 가능)³⁾

- Step Cut 적용 기종 : HS / Bonded

* HS Wafer: Chip Outline to PSV – 35um, Chip Outline to LT – 40um

* NS, TS 기종 Al Pad 사용 가능

※ 소형화(55um) – 0907, 1109(A0, A4), 1411 DPX HTCC, 0907 ~ 1612 PCB (대칭)

Chip Size 기준 1.43 x 1.03 이하

Adv.(Advanced): 고난위도이며, 공정불량 Risk있음

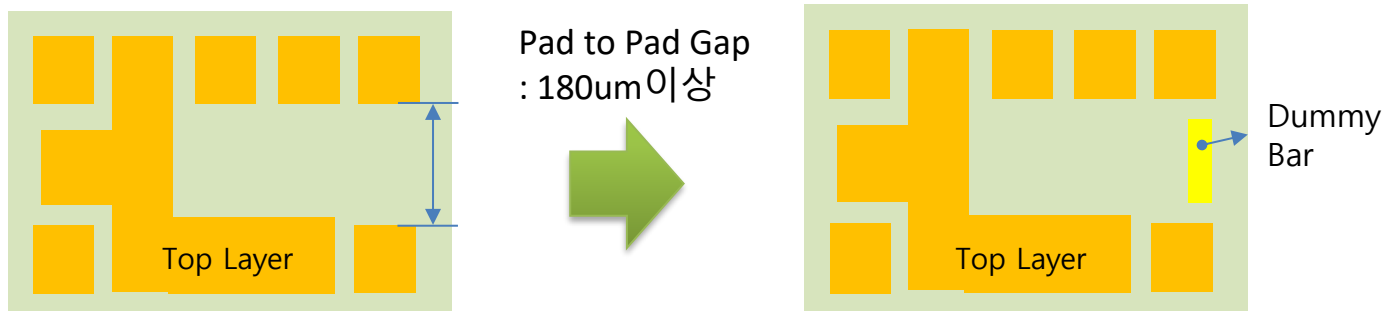
Std.(Standard): 현 PKG Ass'y 공정 능력

Max.(Maximum): 상한 치가 요구될 경우

1) 기판 평탄도 개선 – Metal Pad 추가



2) Epoxy 침투 방지 – Top Layer, Chip Outline위치 Pad to Pad Gap 180um 이상 : Pad 확장 or Dummy Bar Pad 추가 (최소 150 x 50um 면적 이상)

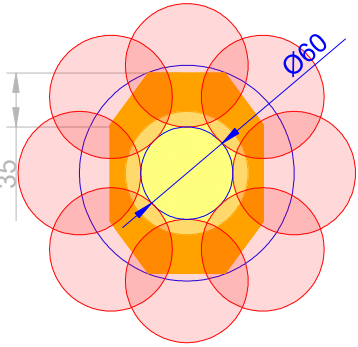
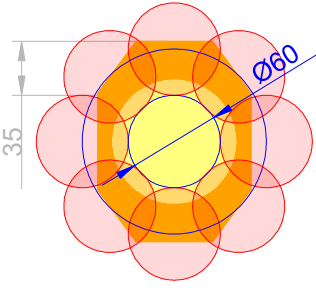
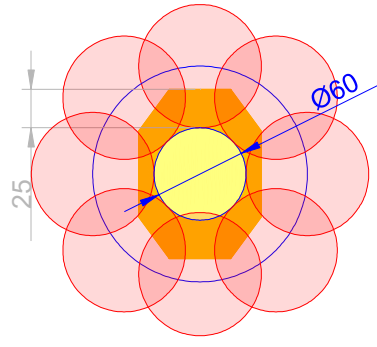
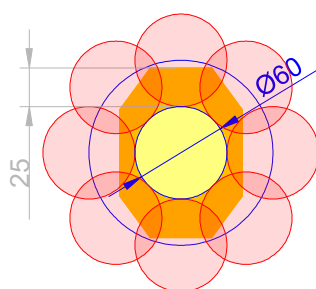


3) Bump 비대칭 점검 구조

HTCC	0907 1109 (A0, A4) 55um Bump	4 Ball	5 Ball	6 Ball			
PCB	1612 (8 Bump) 65um Bump	4 Ball	6 Ball	7 Ball	8 Ball		
	1612 (10 Bump) 65um Bump	5 Ball	6-1 Ball	6-2 Ball	6-3 Ball		
		7-1 Ball	7-2 Ball	8 Ball	9 Ball	10 Ball	

* 비대칭시 Chip to Substrate Top Pad Short Risk 有
: Sort Risk 영역 Top Pad 배선 최소화 必 (XM28ATK)

4) HTCC Bump Pad Via 겹침 : Bump Pad Center 기준 $\Phi 60\mu\text{m}$ 영역 Via 겹침 없을 것

Bump Size	65 μm		55 μm	
Bump Pad Size	130 x 100 μm 		110 x 80 μm 	
Via Size	$\Phi 80\mu\text{m}$ 	$\Phi 60\mu\text{m}$ 	$\Phi 80\mu\text{m}$ 	$\Phi 60\mu\text{m}$ 
기존 Rule 겹침 기준	35 μm 겹침 허용	35 μm 겹침 허용	25 μm 겹침 허용	25 μm 허용
Drawing				
개정 Rule Pad to Via Center	70 μm 이상	60 μm 이상	70 μm 이상	60 μm 이상